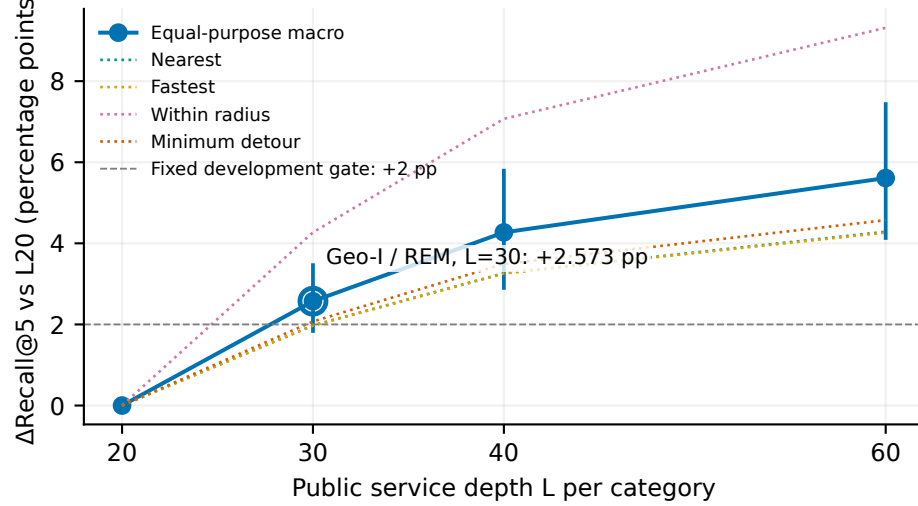
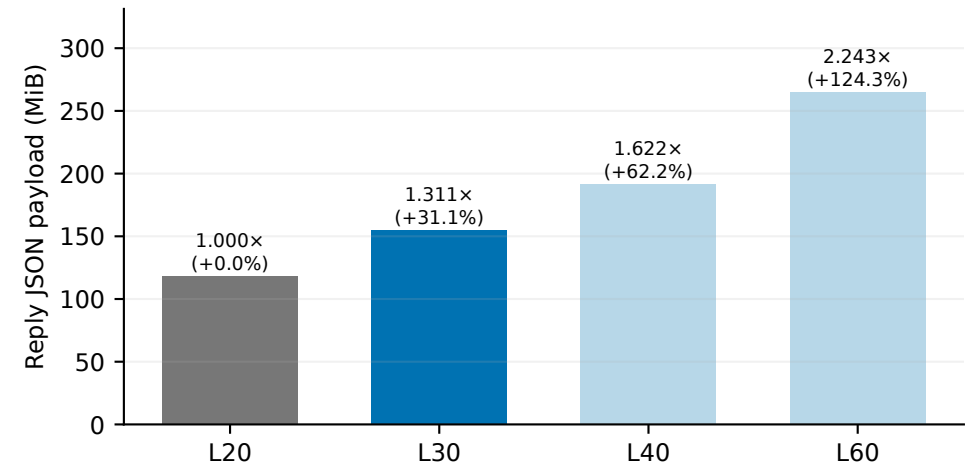


Geo-I / REM: fixed Q, public response-depth utility / cost

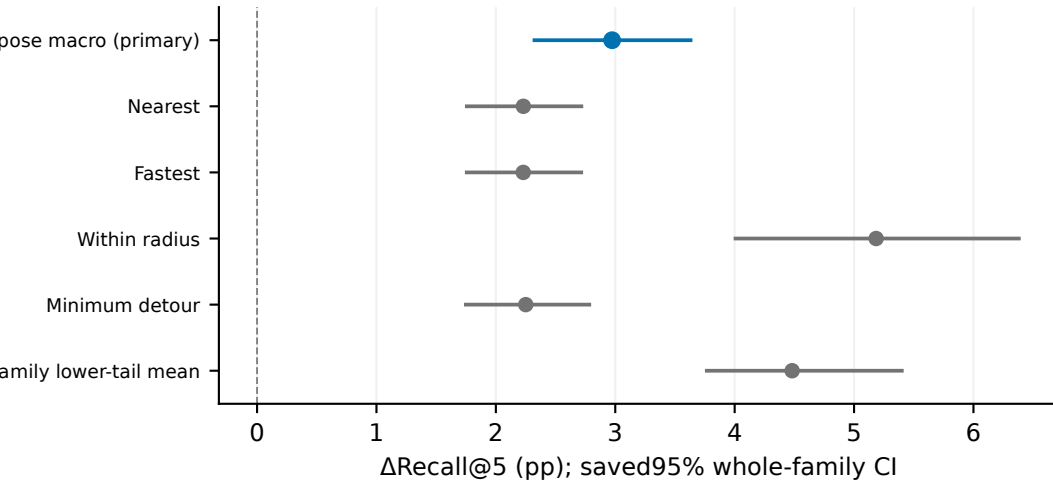
A DEVELOPMENT utility: 6 families
Saved paired-family 95% macro CI; exploratory



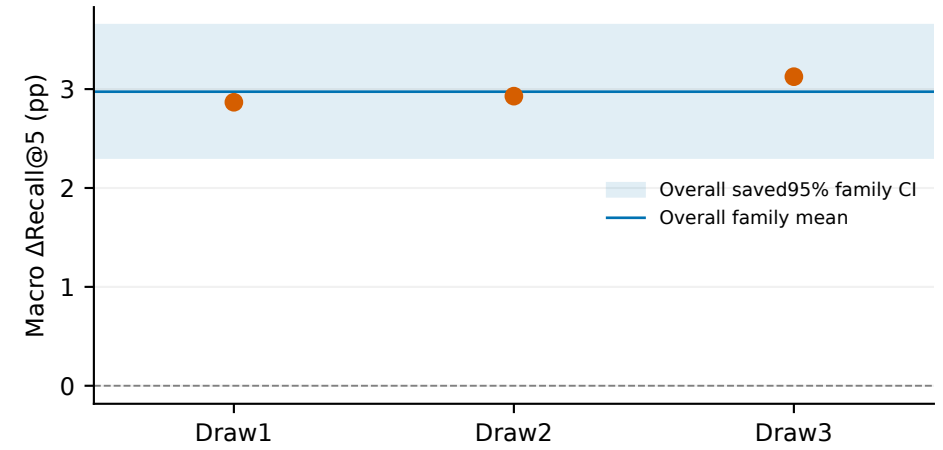
B DEVELOPMENT response cost
Same Q/clock/request count; additional bytes paid



C Fresh Geo-I / REM, L=30 vs L=20
Current replies; same-map synthetic primary



D Nested private-draw diagnostics
Fresh reply JSON: +31.1% vs L20



E Purpose utility and reference coverage: empty reference stays N/A

Purpose	DEV L20 Recall	DEV L30 Recall	DEV reference coverage	Fresh reference coverage
Nearest	94.14%	96.12%	27,216/27,216 (100.0%) windows 4,536/4,536	108,684/108,684 (100.0%) windows 18,114/18,114
Fastest	94.16%	96.13%	27,216/27,216 (100.0%) windows 4,536/4,536	108,684/108,684 (100.0%) windows 18,114/18,114
Within radius	86.60%	90.87%	10,008/27,216 (36.8%) windows 4,506/4,536	42,258/108,684 (38.9%) windows 17,577/18,114
Minimum detour	93.85%	95.92%	27,216/27,216 (100.0%) windows 4,536/4,536	108,684/108,684 (100.0%) windows 18,114/18,114